John M. Dusel, Ph.D.

CONTACT INFORMATION

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EDUCATION

Ph.D. Mathematics (2015), *University of California, Riverside*

M.S. Mathematics (2010), *University of California, Riverside*

B.A. Mathematics, minor Computer Science (2006), *San Diego State University*

Digital Signal Processing Specialization Certificate (2024), *École Polytechnique Fédérale de Lausanne, through Coursera*

EMPLOYMENT HISTORY

- **Senior Data Scientist** (2025 – present)
  Blue Rose Research, New York, NY
- **Adjunct Professor of Computer Science** (2025 – present)
  Frederick Community College, Frederick, Maryland
- **Research Scientist** (2019 – 2025)
  Advanced Data Analytics Division, Metron, Inc., Reston, Virginia
- **Visiting Assistant Professor of Mathematics** (2017 – 2019)
  Department of Mathematics and Computer Science, Mount St. Mary's University (Emmitsburg, Maryland)
- **Visiting Assistant Professor of Mathematics** (2015 – 2017)
  Department of Mathematics and Computer Science, Whittier College
- **Doctoral Student** (2008 – 2015)
  Department of Mathematics, University of California, Riverside
- **Faculty Services Coordinator** (2008)
  Department of Mathematics, University of California, San DIego
- **Doctoral Student** (2007 – 2008)
  Department of Mathematics, University of Florida
- **Computer Resource Specialist** & Webmaster (2006 – 2007)
  Department of Radiology, University of California, San Diego School of Medicine

SCHOLARSHIP, RESEARCH, & DEVELOPMENT

Areas of interest

- Bayesian methodology: Data analysis *incl.* hypothesis testing, prior elicitation, hierarchical modeling, sequential inference, experimental design for testing & evaluation, quality assurance and technical reporting.
- Machine learning: Clustering, deep learning for regression and classification problems, time series forecasting.
- Applied mathematics: Decision theory, digital signal processing, computational geometry, complex adaptive systems. Mathematical modeling in cognitive psychology.
- Pure mathematics: Algebraic combinatorics, representation theory of Lie algebras and connections between category theory and systems dynamics.

Planned/in progress projects

- Bayesian Cost Aware Sequential Probability Ratio Test (joint with L. D. Stone).
- `MaDM-Bayes`: A Python library for Bayesian data analysis for psychological studies. (Collaboration with the Memory and Decision Making Lab at Hood College.) Source code available on GitHub.
- Instruction-Based Inferences in Eyewitness Identification (joint with M. B. Moreland).
- End-to-end software system for experimental data collection and storage, ETL, analysis, and reporting. Planning and initial prototype phase.

Submitted articles

- M. B. Moreland, **J. M. Dusel**, Speed and Accuracy Instructions Invert Effects of Stimulus Class on 2AFC Recognition. Source code for available on GitHub.

Published articles

- J. D. Foley, S. Breiner, E. Subrahmanian, **J. M. Dusel**, Operads for complex system design specification, analysis and synthesis. Proceedings of the Royal Society A., 2021. Available on ResearchGate.
- **J. M. Dusel**, Structural folding and multi-highest-weight subcrystals of $B(\infty)$. Algebras and Representation Theory, 2019. Available on ResearchGate.
- **J. M. Dusel**, Balanced parabolic quotients and branching rules for Demazure crystals. Journal of Algebraic Combinatorics, 2016. Available on ResearchGate.

Open-source software

- SciPy (2025), Matrix variate t distribution for `stats` module.
- `BernoulliBall` (2025), A web app for maintaining and visualizing an estimate of the probability of success for Bernoulli trials. You can also evaluate this estimate against a performance requirement. Source code available on GitHub.

Presentations

- **J. M. Dusel**, John D. Foley, Community-based Modeling with Compositional Foundations. Finalist brief to Metron Innovation Fund Committee, 2023.
- James P. Ferry, **J. M. Dusel**, Ian Herbert, AI/ML-based Motion Prediction. Metron Brown Bag series, 2023.
- John D. Foley, **J. M. Dusel**, Radio Frequency Signal Correlation: ML Technology for tactical SIGINT. Metron Inc. Annual R&D Conference, 2023.
- **J. M. Dusel**, Combinatorial generation of multi-highest-weight crystals. Iowa State University Representation Theory Seminar, 2017.
- **J. M. Dusel**, Crystal folding. Joint Mathematics Meetings, Atlanta, Georgia, 2017.

- **J. M. Dusel**, Balanced parabolic quotients and branching rules for Demazure crystals. American Mathematical Society Western Sectional Meeting, University of Nevada at Las Vegas, 2015.
- **J. M. Dusel**, A self-similar forest and the symmetric group. Mathematics Colloquium, California State University, San Bernardino, 2015.
- **J. M. Dusel**, Self-similar forests and Fibonacci numbers. Mathematical Association of America, Southern California-Nevada Sectional Meeting, Pomona College, 2014

Teaching Experience

Evaluation means in the Table 1 are based on all questions from the "Professor" section of the evaluation form.

Institution	Course	# Sections	Mean Eval.
FCC	Program. Meth. & Obj. Design	1	In progress
MSMU	Mathematical Thought	9	4.1/5
MSMU	Integral Calculus	3	4.1/5
MSMU	Multivariable Calculus	1	4.4/5
MSMU	Geometry	1	4.4/5
Whittier	Elementary Statistics	8	4.6/5
Whittier	Differential Calculus	3	4.7/5
Whittier	Integral Calculus	1	4.8/5
Whittier	Precalculus	1	4.4/5
Whittier	Linear Algebra	1	5.0/5
UCR	Differential Calculus	3	4.5/5
UCR	Integral Calculus	2	4.4/5
UCR	Discrete Math	1	4.5/5
UCR	Precalculus	1	4.5/5
UCR	Algebra Qual Seminar	2	N/A

Table 1: Summary of courses taught as instructor-of-record.

Professional Service

- Curriculum development for Programming Methods and Object Design course. Computer Science program, Frederick Community College, 2025.
  - Contribute new lessons and projects to departmental course notes.
  - Lead planning and integration of standardized development environments (Dev Containers) into curriculum
  - Create documentation to support using GitHub for assignment submission.
  - Develop main repository to be forked and developed on by students through the term. Source code available on GitHub
- Editorial Board Member, PUMP Journal of Undergraduate Research, 2015 – present.
  - Oversee the review process for submitted papers. Select referees, ensure timely turnaround of their reviews, evaluate reviews. Make the final decision to accept or reject. Write and summarize constructive feedback for student authors.

- Diversity, Equity, and Inclusion Committee Member, Metron, Inc., 2014 – 2024.
  - Proposed policies and actions aimed to increase equity for remote employees, by identifying and removing office-centricity and proximity bias at the management and technical team levels. Advocated for more emphasis on recruitment of differently abled candidates.
- Led an independent study course in mathematical logic for senior Philosophy students. Mount St. Mary's University, 2021.
- Advisor for student chapter of the Mathematical Association of America. Mount St. Mary's University, 2020 – 2021.
- Advisor and member of grading committee for senior mathematics theses. Whittier College, 2015 – 2017.
- Search committee member for Visiting Assistant Professor position. Whittier College, 2015 – 2017.
- Chair of Contributed Paper Session on Associative and Nonassociative Rings and Algebras, Category Theory, and Homological Algebra, Joint Mathematica Meetings, 2017.

Industry Experience

- Senior Data Scientist, Blue Rose Research.

  Job description: Collaborate with Data Scientists to design, develop, and implement model training and evaluation workflows. Provide flexible, scalable, and cost-effective infrastructure enabling experimentation and accelerating deployment of production models. Work on all parts of the data processing stack, from initial ingestion and cleaning through model training, evaluation, and inference. Build tools to empower data analyses to derive insights from model outputs.
- Research Scientist, Metron, Inc.

  Job description: Develop mathematical models and algorithms. Develop and demonstrate prototype software, and participate in operational analysis related to technical work. Interact with clients, gather requirements and data. Write and present memoranda and briefings, and contribute to the development of Metron tools. Work in small interdisciplinary teams from the initial system concept, through software and algorithm design, to implementation and demonstration.

  - Task lead for Operational Suitability feature in Metron's Dynamo decision-aid for Bayesian testing & evaluation of complex systems. Collaborated with military testing and evaluation subject-matter experts to develop scenarios, metrics, and models.
  - Lead software developer for a multi-year project on AI/ML for tactical signals intelligence. Responsible party for software demos/deliverables to stakeholders, and for integration into a U.S. Government owned software framework. Managed a team of research scientists and engineers.
  - Individual contributor for algorithm development and user-interface software engineering for interal R&D projects. Topics include computational geometry and Bayesian decision theory applied to distributed control of complex adaptive systems.

Technical Skills

- Programming languages: Python, R, Java, Javascript/Typescript, Mathematica
- Technical computing: Scientific Python ecosystem, PyMC, PyTorch, JAGS, STAN, software-defined radio (GNU Radio, REDHAWK/TOA).
- Data Handling/Manipulation: PostgreSQL, SQLite, SQLalchemy, Pandas.
- Software development: Flask, FastAPI, Vite, Node, React, Docker, Git

Awards, Grants, Etc.

- Grit award for "demonstrating impressive persistence to 'do what must be done' to get a software defined radio (SDR) system up and running for a field demonstration ... and leading the way to timely impact.". Metron, Inc., 2021.
- Faculty research grant. Whittier College, 2016.
- Selected by UCR Mathematics department faculty to serve as Mentor Teaching Assistant, 2014.
- University Teaching Certificate (2012). Program to prepare graduate students for teaching duties in an academic career.
- Awards/fellowships/scholarships during doctoral studies: J. W. and Ida M. Jameson Scholarship (2014), Kramer Memorial Service Award (2014), Graduate Research Mentorship Fellowship (2011 – 2012), Gordon E. Hein Scholarship (2009 – 2012), Outstanding Teaching Assistant Award (2011), Graduate Division Fellowship (2008 – 2010)